Preventive Veterinary Medicine 148 (2017) 89–93

Contents lists available at ScienceDirect

Preventive Veterinary Medicine

journal homepage: www.elsevier.com/locate/prevetmed

Monitoring breeding herd production data to detect PRRSV outbreaks

G.S. Silva[a,b], M. Schwartz[c], R.B. Morrison[d], D.C.L. Linhares[a,*]

[a] Veterinary Diagnostic and Production Animal Medicine Department, Iowa State University, Ames, IA, United States
[b] Veterinary Epidemiology Laboratory, Federal University of Rio Grande do Sul, Porto Alegre, Brazil
[c] Schwartz Farms, Inc., Sleepy Eye, MN, United States
[d] University of Minnesota, St. Paul, Minnesota, United States

ARTICLE INFO

Keywords:
Monitoring
EWMA
SPC
Productivity
Disease detection
Production data
PRRSv

ABSTRACT

Porcine reproductive and respiratory syndrome virus (PRRSv) causes substantial economic impact due to significant losses in productivity. Thus, measuring changes in farm productivity before and after PRRS infection enables quantifying the production and economic impact of outbreaks. This study assessed the application of exponentially weighted moving average (EWMA), a statistical process control method, on selected production data (number of abortions, pre-weaning mortality rate and prenatal losses) to supplement PRRS surveillance programs by detecting significant deviations on productivity in a production system with 55,000 sows in 14 breed-to-wean herds in Minnesota, U.S.A. Weekly data from diagnostic monitoring program (available through the Morrison's Swine Health Monitoring Project) implemented on the same herds was used as reference for PRRS status. The time-to-detect, percentage of early detection of PRRSv-associated productivity deviations, and relative sensitivity and specificity of the production data monitoring system were determined relative to the MSHMP. The time-to-detect deviations on productivity associated with PRRS outbreaks using the EWMA method was −4 to −1 weeks (interquartile range) for the number of abortions, 0–0 for preweaning mortality and −1 to 3 weeks for prenatal losses compared to the date it was reported in the MSHMP database. Overall, the models had high relative sensitivity (range 85.7–100%) and specificity (range 98.5%–99.6%) when comparing to the changes in PRRS status reported in the MSHMP database. In summary, the use of systematic data monitoring showed a high concordance compared to the MSHMP-reported outbreaks indicating that on-farm staff and veterinary oversight were efficient to detect PRRSv, but can be more efficient if they were monitoring closely the frequency of abortions. The systematic monitoring of production indicators using EWMA offers opportunity to standardize and semi-automate the detection of deviations on productivity associated with PRRS infection, offering opportunity to early detect outbreaks and/or to quantify the production losses attributed to PRRS infection.

1. Introduction

Porcine reproductive and respiratory syndrome (PRRS) was first reported in the USA swine industry in 1989 (Keffaber, 1989) and remains a challenge causing substantial economic losses due to significant changes in productivity parameters (Holtkamp et al., 2013). Although the apparent incidence in breeding herds has declined in recent years (Morrison et al., 2015), the prevalence continues to increase over time (SHMP, 2017), which may be related to increasing attempts to control rather than eliminate PRRS from breeding herds.

One of the objectives of ongoing surveillance of livestock operations is to early detect pathogen introduction allowing rapid response by implementing measures to control the infection and/or prevent spread to other sites (Salman, 2013). Conventional PRRS monitoring systems include routine diagnostic testing for PRRSv nucleic acid (by PCR based assays) and/or for anti-PRRSv-antibodies (by serology) using biological samples including serum samples (Dewey and Straw, 2006) or oral fluids (Prickett et al., 2008). Efficient surveillance enabling early detection of PRRS in breeding herds may contribute to reduce PRRS incidence by adopting practices to prevent direct and or indirect transmission to other herds at risk.

Since PRRSv infection in breeding herds is characterized by significant changes in productivity including increased abortions, preweaning mortality and neonatal losses, measuring significant deviations in production records would enable assessing the production and economic impact (De Vries and Reneau, 2010). The applicability of monitoring production indicators or animal behavior has been described as a proxy of animal health or disorder (Weary et al., 2009).

* Corresponding author at: (DCLL) 2221 Lloyd Veterinary Medical Center. 1600 16S Street. Ames, IA, 50 011, United States.
E-mail address: linhares@iastate.edu (D.C.L. Linhares).

http://dx.doi.org/10.1016/j.prevetmed.2017.10.012
Received 15 February 2017; Received in revised form 18 October 2017; Accepted 18 October 2017
0167-5877/ © 2017 Elsevier B.V. All rights reserved.

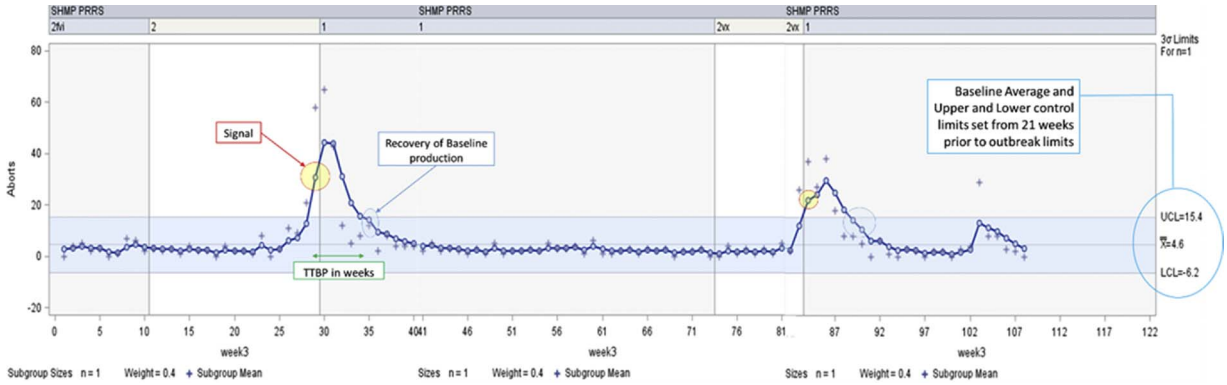

Fig. 1. Example of detecting significant deviation in abortions and change in PRRS status over time. The change of PRRS status over time is described on the top section of the figure: status 2fvi means stable breeding herds with ongoing field virus inoculation, 2 means stable herds and 2vx means stable herds with ongoing exposure of breeding females to a modified live virus vaccine. Changes to status 1 means that the herd had a PRRS outbreak. A signal represents a point of the smoothed line (blue) out of the upper control limit. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Statistical process control (SPC) charts are widely used to evaluate variations in industrial processes (Montgomery, 2009), health-care and public-health surveillance (Woodall and Tech, 2006). Within agriculture, they have been described for dairy production (de Vries and Conlin, 2003; Lukas et al., 2005; Reneau and Kinsel, 2001). There is still only a limited number of reports of applying SPC tools to swine herds to measure production losses due to diseases (Cornou et al., 2008; Engler et al., 2009) and some of these applications were based on simulation data (Kirchner et al., 2006; Krieter et al., 2009). Thus, the implementation of SPC can provide helpful orientation to managers by monitoring the patterns of production and potential disease outbreaks.

The objective of this study was to assess the feasibility of systematically monitoring key production indicators using a SPC method to detect PRRS outbreaks in breed-to-wean pig herds. Furthermore, we assessed the effectiveness of SPC to detect of PRRSv outbreaks in relation to the MSHMP, a voluntary disease reporting program that monitors changes in PRRS status in US breeding herds.

2. Material and methods

2.1. Study design

This was a retrospective observational study that used selected key production indicators and pre-defined SPC analysis to detect significant deviations from herd-specific baseline. Deviations (signals) were compared to herd PRRS status by week, which was based on routine diagnostic monitoring. We described the relative sensitivity, specificity and time-to-detect outbreaks using a SPC method, as compared to documented PRRS outbreaks reported to the MSHMP.

2.2. PRRSv status, eligibility criteria and source population

The Morrison's Swine Health Monitoring Project (MSHMP) is a network of U.S.A. swine producers. MSHMP represents a convenience sample of 910 sow herds including a total of approximately 2.3 million sows. Veterinarians represent each herd and voluntarily report diagnostic status for PRRSV and PEDV on a weekly basis (Goede and Morrison, 2016; Tousignant et al., 2015). Diagnostic records were not audited and veterinarians' reports were considered correct. This included the diagnosis of PRRSV and the status of the herd for the disease following herd classification guidelines described previously (Holtkamp et al., 2011). According to MSHMP, a PRRS outbreak is confirmed by positive PCR and virus isolation, which is required after farmers detected signals suggesting virus circulation (e.g. clinical signs, production data or routine monitoring).

To address our objective, we sought production system from the MSHMP system fulfilling the following eligibility criteria: (a) availability of weekly PRRS and PED status; (b) willingness of producer to share weekly production data; (c) continuous breed-to-wean operation; (d) herd having at least 21 weeks with no PRRS and PED outbreak to create baseline References

A production system with 55,000 sows consisting of 14 breeding herds located in Minnesota, US was enrolled in the study. The weekly production data was extracted from a commercial record keeping system (PigCHAMP 4.5.4, PigCHAMP Inc, Ames, IA). The final database had 108 weeks per herd (1512 herd-weeks) and weekly PRRSV.

2.3. Production indicators

The parameters were selected based on the most common clinical signs on herds affected by PRRSV infection (Pozzi and Alborali, 2012; Zimmerman et al., 2012). More specifically, significant changes were monitored on a week basis for number of abortions, pre-weaning mortality rate (PWM) and prenatal losses (difference between 'total number of pigs born per litter' and 'number of pigs born alive per litter').

2.4. Statistical process control method

Exponentially weighted moving average (EWMA) was the SPC method selected to implement and measure the outcomes. The chart parameters were set with the intent to minimize false-alarm rate (high specificity) and detect large shifts, thus the sigma (σ) was set to 3 and smoothing parameter (λ) was set to 0.40. It is not the intention of the study evaluate different parameters to test is validity, and the chart parameters we test was already used (Goede et al., 2015; Krieter et al., 2009; Linhares et al., 2014). A chart example can be seen in Fig. 1. The models were implemented using Proc MACONTROL from SAS version 9.4 software (SAS Institute Inc., Cary, North Carolina).

2.5. Outcomes

The outcomes used to report effectiveness of SPC analysis compared to conventional routine diagnostic monitoring of PRRS included: percentage of early detection of PRRS outbreaks, time-to-detect (TTD) PRRSV outbreak and time to recover baseline production (TTBP). These outcomes were calculated for each production parameter in each herd monitored (n = 14) herd, and the median results, along with interquartile ranges are reported. Early detection was defined as detecting a significant SPC signal at least 1 week before a PRRS outbreak was recorded in the MSHMP database. TTD (also known as average time to signal (De Vries and Reneau, 2010)) was defined as the difference in weeks between the change of MSHMP PRRS status and the detection of a SPC signal. TTBP represented the number of weeks that the herd took

Table 1
Contingency table to access the performance the SPC method for each parameter compared to reported PRRS outbreaks.

		Status of the parameter to be evaluated	
MSHMP-reported PRRSV outbreak (reference)		Increased	Normal
	Yes	TP	FN (type II error)
	No	FP (type I error)	TN

TP: true positive; FN: false negative; FP: false positive; TN: true negative.

to recover the same level of productivity that it had before the outbreak.

2.6. Model performance

The performance of the production indicators using EWMA method was examined using contingency tables to compare the model results relative to the PRRSV outbreaks reported to MSHMP, which was considered the gold standard (Table 1). Thus relative sensitivity, relative specificity, number of false alarms, false positive rate, false negative rate, and accuracy were calculated for each production parameter using established methods to validate diagnostic assays (McKenna and Dohoo, 2006) and SPC methods (Quimby et al., 2001). For each production data, we evaluated the overall performance using the sum of all weeks monitored in the participating herds. Each week was classified if it had an outbreak reported in MSHMP database (status 1 − Yes/No) and if EWMA detected or not a signal (Yes/No) (Table 1). The relative performance was calculated using the package epiR in software R. Accuracy here is defined as the proportion of correctly classified weeks among all weeks under observation [(TP + TN)/(TP + TN + FP + FN)].

3. Results

The production system reported 10 PRRS outbreaks during the observation period. Two herds had 2 outbreaks each, 6 herds had one outbreak and 6 herds reported no outbreaks. All 14 herds fulfilled the eligibility criteria, contributing 108 herd-weeks for "aborts" and "preweaning mortality" and 102 herd-weeks for prenatal losses. Two herds did not meet the eligibility criteria for prenatal losses due to not having at least 21 weeks of within control limits (3 sigma) to create a baseline and were removed for that specific parameter.

Early detection rate was higher for abortions (90%) compared with prenatal losses (57%) and PWM (10%) (Table 2). The interquartile range (25th to 75th percentile) of time to detect (TTD) outbreak varied from −4 to 3 weeks among the measured parameters (Fig. 2). Abortion had the earliest TTD, detecting outbreaks up to 4 weeks prior to the reported date in the MSHMP database. The last indicator to detect PRRSV outbreak was prenatal losses (median −1.00, IQ range of −1 to 3).

Table 2
Early detection rate, average time to detect (TTD) PRRS outbreak and time to baseline production (TTBP) of each production parameter.

Production parameters	Early Detection (%)	TTD median (weeks)	TTD IQ Range (weeks)	TTBP median (weeks)	TTBP IQ Range (weeks)
Abortions	90% (9/10)	−1.00	−4 to −1	8.00	5 to 11.25
PWM	10% (1/10)	0.00	0 to 0	9.00	2.5 to 16.50
Prenatal losses	57% (4/7)	−1.00	−1 to 3	16.00	7 to 30

Early detection – number of OB detected before or same time as the change on PRRSV status (SHMP); TTD – Time-to-detect outbreak; IQ Range – Interquartile range (25th to 75th percentile); TTBP – Time-to-baseline production; PWM – Preweaning mortality.

The median TTBP for abortions, PWM and prenatal losses was 8.0, 9.0 and 16.0 weeks, respectively (Table 2). The IQ ranged from 2.5 weeks (PWM, lowest) to 30 weeks (prenatal losses, highest).

The relative performance for each parameter using the SPC method to detect production parameter signals achieved good performance to identify PRRS outbreaks reported to the MSHMP system (Table 3). Monitoring abortions and PWM were better than prenatal losses to identify signals associated with PRRS outbreaks. Likewise, the number of false detection events monitoring prenatal losses was twice as high as that of abortions and PWM.

4. Discussion

The systematic monitoring of weekly production indicators allowed detection of deviations on productivity that were closely associated with PRRS outbreaks. Continuous evaluation of animal health is essential for sustainable production (Weary et al., 2009), allowing early detection of deviations, and subsequently quick implementation of applicable intervention methods. The production indicators used in this study were chosen based on their biological relevance to PRRSV infection in swine breeding herds (Engler et al., 2009; Pozzi and Alborali, 2012; Zimmerman et al., 2012). Monitoring production data was useful to detect signals associated with PRRSV infection, but in daily production this tool also could assist the identification of non-infectious sources of variation or another infectious disease, since production indicators are not disease-specific.

The benefit of detecting signals (of animal health, production indicators, or behavior metrics) potentially associated with disease outbreaks include the early confirmation of infection with diagnostic tests, which in turn allows adopting rapid response practices to mitigate the infection in the herd and/or to prevent the spread of pathogens among farms sharing resources with a larger network (e.g. sharing vehicles, pig flow, people flow and/or supplies among a production system). In this specific case, the systematic monitoring of production indicators by the MSHMP participants could serve as a trigger to anticipate the PRRSV ongoing surveillance system given the identification of clinical signs of the disease.

Based on aborts, the median TTD outbreak was − 1.0 weeks (25th to 75th percentile range −4 to −1). Another study (Engler et al., 2009) reported similar results monitoring PWM and identified production losses 4 and 3 weeks before PRRSV was diagnosed in 2 farms respectively. Similar results were not obtained using PWM in this study, but the consistent finding of TTD of 0.0 weeks suggests that the production system was already using this parameter as a key indicator of possible PRRSv circulation, and following up with diagnostic investigation to confirm the outbreak. The performance of the model using prenatal losses was inferior to the number of aborts or PWM.

The TTBP is a measure that indicates the duration of production losses associated with disease outbreaks. In the literature only a few studies demonstrated the direct benefit/implication for producers to detect deviations on productivity related to disease occurrence using SPC methods (Engler et al., 2009; Madsen and Kristensen, 2005; Quimby et al., 2001). We reported a shorter median, and lower variation in TTBP when compared to studies reporting TTBP for number of pigs weaned per week following PRRS or PED outbreaks (Goede and Morrison, 2016; Linhares et al., 2014). Pig farms usually have targets for certain production indicators, here, in addition to showing the application of the method to monitor signs of disease, the method was also applied to estimate production losses (TTBP). The same method can be used by farmers to estimate the cost-benefit of interventions (e.g. health, management) at herd-level.

De Vries et al. reviewed studies applying SPC methods in animal production and stressed that there was limited information on model performance (Se, Sp, FPR, FNR) due to the lack of comparison with a reference and/or gold standard measure of disease presence (De Vries and Reneau, 2010). In this study, we assumed that the MSHMP PRRS

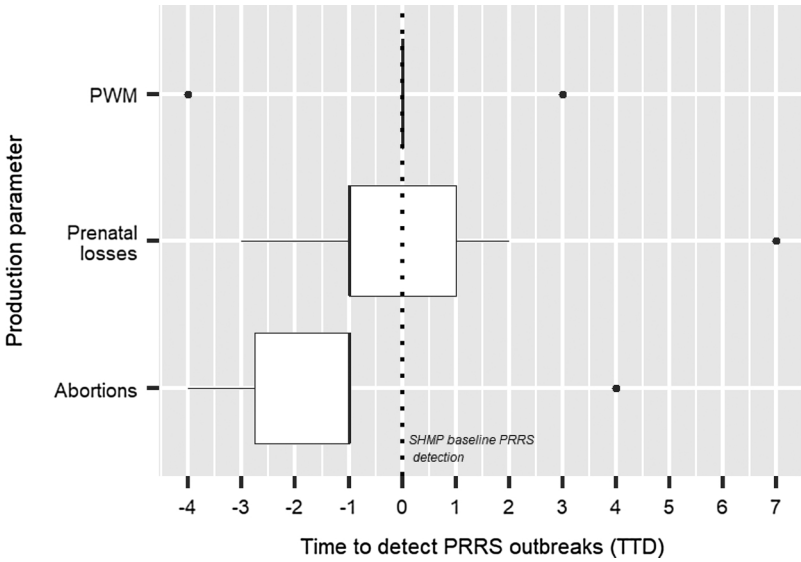

Fig. 2. Distribution of the time-to-detect PRRS outbreaks for each production parameter. The time-to-detect the SPC can be compared to the SHMP status (time 0), when the herd detected PRRSV using routine diagnostic test. This suggests that the diagnoses could have been made before and the SPC could have been used to aid in the timing of the test.

Table 3
Performance of the production indicators using EWMA model to detect significant variations in productivity associated with reported PRRS outbreaks. The performance was determined relatively to MSHMP PRRSV status database.

Production parameters	Weeks monitored (n)	TP	FP	FN	TN	Relative Se	95% CI	Relative Sp	95% CI	FPR − type I error	FNR − type II error	Relative ACC
Abortions	1398	10	7	0	1381	100.0%	[69.0–100.0]	99.5%	[99.1–99.8]	0.5%	0.0%	99.5%
PWM	1368	10	5	0	1353	100.0%	[69.0–100.0]	99.6%	[99.3–99.9]	0.4%	0.0%	99.6%
Prenatal losses	1026	6	15	1	1004	85.7%	[42.0–100.0]	98.5%	[97.7–99.3]	1.5%	14.3%	98.4%

TP – true positive; FP – false positive; FN – false negative; TN – true negative.
Se – sensitivity; Sp – specificity; FPR – False Positive Rate FPR; FNR – False Negative Rate.
Accuracy – ACC.

status was the gold standard, reflecting the true disease status to compare the performance of the SPC analyses. Thus, the performance values of the model were relative to the MSHMP and should be interpreted with caution. The high relative sensitivity and specificity obtained can be due to the low number of PRRSV outbreaks (n = 10) during the investigated period, and the high number of negative weeks monitored. Likewise, it is possible that some minor PRRS outbreak were not detected and never reported to the MSHMP. We selected EWMA as the SPC method due to the great sensitivity and specificity to detect shifts in processes when compared to other methods (Mertens et al., 2011). Moreover, EWMA is easily automated, consistent, and does not require normal-distributed data.

To calculate the relative performance of the production indicators, we considered the aggregated data of all participating herd-weeks. We used one set of EWMA parameters (0.4 lambda and 3.0 sigma) previously reported to detect high variations of swine production data at a low false alarm rate (Goede and Morrison, 2016; Krieter et al., 2009; Linhares et al., 2014; Mertens et al., 2011). Even trying to minimize the risk of making a type I error using sigma equal to 3 for the control limits (Montgomery, 2009), the majority of the classification errors obtained in the 14 herds were FP showing the difficulty of monitoring biological processes. The FP signals may be associated with either failure to record production data, a non-PRRS infectious cause, a non-infectious cause (e.g. employee changes, management changes) or method failure. The greater the number of false alarms, the lesser is the credibility of the method for producers. This drawback can be handled by monitoring multiple indicators at the same time, as demonstrated, improving the reliability of the monitoring process by potentially investigating the false alarms by comparing the monitored indicators. Given the results of the study, monitoring the number of abortions and PWM had the lowest error rates.

A limitation of this study is on the assumption that the MSHMP PRRSv was a gold standard for defining PRRS outbreaks and the PRRS status provided by participants is assumed to be true. Therefore, we report the 'relative' model performance, instead of performance. Also, due to the results obtained, we did not explore the effect of variation of model parameters (sigma and lambda) on the relative performance, which could identify the set of parameters that minimize classification errors (FP and FN).

We demonstrated the feasibility of the application of a EWMA method to monitor productivity, identifying signals potentially associated with infectious disease introduction. Diagnostic investigation should be implemented to further characterize the sources of variations detected with SPC methods. Monitoring multiple parameters may increase the reliability to early detect and/or to measure the impact of infectious diseases in livestock operations, when compared to monitoring a single parameter.

In conclusion, under the conditions of this study, systematic monitoring of key production indicators resulted in detection of signals that were closely associated with PRRS outbreaks reported to the MSHMP. The number of abortion was the most efficient parameter, detecting outbreaks up to 4 weeks before it was recorded in the MSHMP database. MSHMP participants and swine producers can take advantage of monitoring production data to enrich on-going surveillance program. Results support the establishment of an on-going monitoring of production data as part of disease surveillance to detect endemic, and/or emerging pathogen introduction in production systems.

Acknowledgments

This study was funded by the Iowa State University Veterinary Diagnostic and Production Animal Medicine Department. We thank the

Schwartz Farms, Inc. to collaborate by providing the production data used in the analyses. We also acknowledge the critical review this manuscript by Dr. Luis Gustavo Corbellini. Moreover, we thank two anonymous Preventive Veterinary Medicine reviewers for their critical comments on this manuscript.

References

Cornou, C., Vinther, J., Kristensen, A.R., 2008. Automatic detection of oestrus and health disorders using data from electronic sow feeders. Livest. Sci. 118, 262–271. http://dx.doi.org/10.1016/j.livsci.2008.02.004.

De Vries, A., Reneau, J.K., 2010. Application of statistical process control charts to monitor changes in animal production systems. J. Anim. Sci. 88, 11–24. http://dx.doi.org/10.2527/jas.2009-2622.

Dewey, C.E., Straw, B.E., 2006. Herd evaluation. In: Straw, B.E., Zimmerman, J.J., D'Allaire, S., Taylor, D.J. (Eds.), Diseases of Swine. Blackwell Publishing, Ames, IA, pp. 5–17.

Engler, J., Tölle, K.-H., Timm, H.H., Hohls, E., Krieter, J., 2009. Control charts applied to pig farming data. Arch. Tierzucht 52, 272–283.

Goede, D., Morrison, R.B., 2016. Production impact & time to stability in sow herds infected with porcine epidemic diarrhea virus (PEDV). Prev. Vet. Med. 123, 202–207. http://dx.doi.org/10.1016/j.prevetmed.2015.11.010.

Goede, D., Murtaugh, M.P., Nerem, J., Yeske, P., Rossow, K., Morrison, R., 2015. Previous infection of sows with a mild strain of porcine epidemic diarrhea virus confers protection against infection with a severe strain. Vet. Microbiol. http://dx.doi.org/10.1016/j.vetmic.2014.12.019.

Holtkamp, D.J., Polson, D.D., Torremorell, M., Morrison, B., Classen, D.M., Becton, L., Henry, S., Rodibaugh, M.T., Rowland, R.R., Snelson, H., Straw, B., Yeske, P., Zimmerman, J., 2011. Terminology for classifying the porcine reproductive and respiratory syndrome virus (PRRSV) status of swine herds. J. Swine Health Prod. 19, 44–56 (11020101 [pii]).

Holtkamp, D.J., Kliebenstein, J.B., Neumann, E.J., Zimmerman, J.J., Rotto, H.F., Yoder, T.K., Wang, C., Yeske, P.E., Mowrer, C.L., Haley, C. a., 2013. Assessment of the economic impact of porcine reproductive and respiratory syndrome virus on United States pork producers. J. Swine Health Prod. 21, 72–84.

Keffaber, K.K., 1989. Reproductive failure of unknown etiology. Am. Assoc. Swine Pract. Newsl. 1 (2), 1–10 (1).

Kirchner, K., Tölle, K.H., Krieter, J., 2006. Optimisation of the decision tree technique applied to simulated sow herd datasets. Comput. Electron. Agric. 50, 15–24. http://dx.doi.org/10.1016/j.compag.2005.07.002.

Krieter, J., Engler, J., Tölle, K.H., Timm, H.H., Hohls, E., 2009. Control charts applied to simulated sow herd datasets. Livest. Sci. 121, 281–287. http://dx.doi.org/10.1016/j.livsci.2008.06.025.

Linhares, D.C.L.C.L., Cano, J.P.P., Torremorell, M., Morrison, R.B.B., 2014. Comparison of time to PRRSv-stability and production losses between two exposure programs to control PRRSv in sow herds. Prev. Vet. Med. 116, 111–119. http://dx.doi.org/10.1016/j.prevetmed.2014.05.010.

Lukas, J.M., Hawkins, D.M., Kinsel, M.L., Reneau, J.K., 2005. Bulk tank somatic cell counts analyzed by statistical process control tools to identify and monitor subclinical mastitis incidence. J. Dairy Sci. 88, 3944–3952 https://doi.org/10.3168/jds.S0022-0302(05)73080-0.

Madsen, T.N., Kristensen, A.R., 2005. A model for monitoring the condition of young pigs by their drinking behaviour. Comput. Electron. Agric. 48, 138–154. http://dx.doi.org/10.1016/j.compag.2005.02.014.

McKenna, S.L.B., Dohoo, I.R., 2006. Using and interpreting diagnostic tests. Vet. Clin. North Am. Food Anim. Pract. 22, 195–205. http://dx.doi.org/10.1016/j.cvfa.2005.12.006.

Mertens, K., Decuypere, E., De Baerdemaeker, J., De Ketelaere, B., 2011. Statistical control charts as a support tool for the management of livestock production. J. Agric. Sci. 149, 369–384. http://dx.doi.org/10.1017/S0021859610001164.

Montgomery, D.C., 2009. Introduction to Statistical Quality Control, 7th ed. John Wiley and Sons, New York. John Wiley & Sons, Hoboken, NJ 10.1002/1521-3773(20010316)40:6 < 9823:AID-ANIE9823 > 3.3. CO;2-C.

Morrison, R.B., Goede, D., Tousignant, S., Perez, A., 2015. Current Incidence, Prevalence and State of Control of PRRS Virus in North America. International PRRS Congress, Ghent, Belgium, pp. 2015.

Pozzi, P.S., Alborali, G.L., 2012. Reproductive diseases in sows (Sus scrofa domestica): A review. Isr. J. Vet. Med. 67, 24–33.

Prickett, J.R., Kim, W., Simer, R., Yoon, K.-J., Zimmerman, J., 2008. Oral-fluid samples for surveillance of commercial growing pigs for porcine reproductive and respiratory syndrome virus and porcine circovirus type 2 infections. J. Swine Health Prod. 16, 86–91.

Quimby, W.F., Sowell, B.F., Bowman, J.G.P., Branine, M.E., Hubbert, M.E., Sherwood, H.W., 2001. Application of feeding behaviour to predict morbidity of newly received calves in a commercial feedlot. Can. J. Anim. Sci. 81, 315–320. http://dx.doi.org/10.4141/A00-098.

Reneau, J.K., Kinsel, M.L., 2001. Record systems and herd monitoring in production-oriented health and management programs in food producing animals. In: Radostitis, O. (Ed.), Herd Health. Saunders Company, Philadelphia, PA.

SHMP, 2017. Swine Health Monitoring Project. (1/17/2017).

Salman, M.D., 2013. Surveillance tools and strategies for animal diseases in a shifting climate context. Anim. Heal. Res. Rev. 14, 147–150. http://dx.doi.org/10.1017/S1466252313000121.

Tousignant, S.J.P., Perez, A.M., Lowe, J.F., Yeske, P.E., Morrison, R.B., 2015. Temporal and spatial dynamics of porcine reproductive and respiratory syndrome virus infection in the United States. Am. J. Vet. Res. 76, 70–76. http://dx.doi.org/10.2460/ajvr.76.1.70.

Weary, D.M., Huzzey, J.M., Von Keyserlingk, M.A.G., 2009. Board-invited review: using behavior to predict and identify ill health in animals. J. Anim. Sci. 87, 770–777. http://dx.doi.org/10.2527/jas.2008-1297.

Woodall, W.H., Tech, V., 2006. The use of control charts in surveillance. J. Qual. Technol. 38, 89–104. http://dx.doi.org/10.1111/j.1600-0420.2005.00625.x.

Zimmerman, J.J., Benfield, D., Dee, S.A., Murtaugh, M.P., Stadejek, T., Stevenson, G.W., Torremorell, M., 2012. Porcine reproductive and respiratory syndrome virus (Porcine arterivirus). In: Zimmerman, J.J., Karriker, L.A., Ramirez, A., Schwartz, K.J., Stevenson, G.W. (Eds.), Diseases of Swine. Wiley-Blackwell, pp. 461–486.

de Vries, A., Conlin, B.J., 2003. Design and performance of statistical process control charts applied to estrous detection efficiency. J. Dairy Sci. 86, 1970–1984. http://dx.doi.org/10.3168/jds.S0022-0302(03)73785-0.